

- Xiaohan Xu, Chongyang Tao, Tao Shen, Can Xu, Hongbo Xu, Guodong Long, and Jian-Guang Lou. Re-reading improves reasoning in language models, 2024b. URL <https://openreview.net/forum?id=3jXCF5dNpC>.
- Chengrun Yang, Xuezhi Wang, Yifeng Lu, Hanxiao Liu, Quoc V Le, Denny Zhou, and Xinyun Chen. Large language models as optimizers. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=Bb4VGOWELI>.
- Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L. Griffiths, Yuan Cao, and Karthik Narasimhan. Tree of Thoughts: Deliberate problem solving with large language models, 2023.
- Xi Ye, Qiaochu Chen, Isil Dillig, and Greg Durrett. Satisfiability-aided language models using declarative prompting. In *Advances in Neural Information Processing Systems*, 2023.
- Hugh Zhang, Jeff Da, Dean Lee, Vaughn Robinson, Catherine Wu, Will Song, Tiffany Zhao, Pranav Raja, Dylan Slack, Qin Lyu, Sean Hendryx, Russell Kaplan, Michele Lunati, and Summer Yue. A careful examination of large language model performance on grade school arithmetic. *ArXiv*, abs/2405.00332, 2024. URL <https://api.semanticscholar.org/CorpusID:269484687>.
- Huaixiu Steven Zheng, Swaroop Mishra, Xinyun Chen, Heng-Tze Cheng, Ed H. Chi, Quoc V Le, and Denny Zhou. Take a step back: Evoking reasoning via abstraction in large language models. In *The Twelfth International Conference on Learning Representations*, 2024a. URL <https://openreview.net/forum?id=3bq3jsvcQ1>.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-bench and Chatbot Arena. In *Proceedings of the 37th International Conference on Neural Information Processing Systems*, NIPS ’23, Red Hook, NY, USA, 2024b. Curran Associates Inc.
- Wanjun Zhong, Ruixiang Cui, Yiduo Guo, Yaobo Liang, Shuai Lu, Yanlin Wang, Amin Saied, Weizhu Chen, and Nan Duan. AGIEval: A Human-Centric Benchmark for Evaluating Foundation Models, 2023.
- Ben Zhou, Kyle Richardson, Xiaodong Yu, and Dan Roth. Learning to decompose: Hypothetical question decomposition based on comparable texts. In Yoav Goldberg, Zornitsa Kozareva, and Yue Zhang (eds.), *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pp. 2223–2235, Abu Dhabi, United Arab Emirates, December 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.emnlp-main.142. URL <https://aclanthology.org/2022.emnlp-main.142>.
- Ben Zhou, Hongming Zhang, Sihao Chen, Dian Yu, Hongwei Wang, Baolin Peng, Dan Roth, and Dong Yu. Conceptual and unbiased reasoning in language models. *ArXiv*, abs/2404.00205, 2024. URL <https://api.semanticscholar.org/CorpusID:268820105>.
- Denny Zhou, Nathanael Schärli, Le Hou, Jason Wei, Nathan Scales, Xuezhi Wang, Dale Schuurmans, Claire Cui, Olivier Bousquet, Quoc V Le, and Ed H. Chi. Least-to-most prompting enables complex reasoning in large language models. In *The Eleventh International Conference on Learning Representations*, 2023a. URL <https://openreview.net/forum?id=WZH7099tgfM>.
- Yongchao Zhou, Andrei Ioan Muresanu, Ziwen Han, Keiran Paster, Silviu Pitis, Harris Chan, and Jimmy Ba. Large Language Models are Human-Level Prompt Engineers. In *The Eleventh International Conference on Learning Representations*, 2023b. URL <https://openreview.net/forum?id=92gvk82DE->.

A META-ANALYSIS EXPANDED DETAILS ON CRITERIA AND PROCESS

Automatic Selection and Paper Filtering We investigate all papers from ICLR 2024, a representative ML venue, and two representative NLP venues, EACL 2024 and NAACL 2024 (including Findings and Workshop papers). We filtered all 4,642 papers (2,259 from ICLR 2024 and 2,382 from the two ACL-affiliated conferences) for those with at least two occurrences of “CoT”, “chain-of-thought”, or “chain of thought”, resulting in 516 papers. There are conceivably papers using CoT called by another name (e.g., Scratchpads), but we believe these 516 give a representative sample appropriate for systematic analysis.

Manual Paper Filtering and Results Extraction We then filter down to papers that perform a comparison of CoT prompting vs. direct prompting, whether or not this is core to the paper’s research question. We manually filtered the 516 papers in question and extracted the key results from those that remained. We excluded multimodal models, CoT-fine-tuned models, any experiments where the “CoT” method involves multiple forward passes (e.g., self-consistency (Wang et al., 2023c) and tree-of-thought (Yao et al., 2023)),⁵ and systems that augment LLMs with external tools (discussed more in Section 5).

For each paper passing through these criteria, we manually extracted the results from key tables comparing CoT and direct answer prompts. We only include results where the CoT and direct prompts are run on the same model and same dataset while being on a scale of 0 to 100 (excluding Likert scale evaluations, for example) for a more direct comparison. When papers include various CoT or direct answer prompts (including zero/few-shot variants), we always take the best-performing prompt for both. We focus on key test results where applicable, excluding dev sets if they are reported alongside test and also excluding numbers from ablations or nonstandard subsets of datasets.

This resulted in a total of 1,218 experimental comparisons across 110 papers (35 from ICLR and 75 from NAACL and EACL) covering 264 datasets. Details and more information will be available in our GitHub Repo.

Categorization Given the large number of tasks and datasets being compared, we grouped each task into a set of 14 categories. These categories were determined based on the description (and possibly examples) of the task, not taking into account system performance. These categories abstract over traditional NLP task classifications (e.g., NER, reading comprehension) and take into account both the task format and the kinds of reasoning involved. Definitions for several categories are shown in Table 1 and the full description is given in Appendix B.

B QUANTITATIVE META-ANALYSIS

See the full list of categories and their descriptions that we used for the meta-analysis in Table 2.

C EXPANDED EXPERIMENTAL DETAILS

A full list of the datasets can be found in Table 4. Each model can be seen in Table 5. We use one answer parser for all datasets of the same answer response format (one for multiple choice, short answer, etc.); however, some datasets require special handling and have edge cases that we handle separately from the rest of the datasets. Similarly, for each model, we use the exact same prompt across them, except when closed source models require different prompts because they do not allow for partial completions (i.e., when we cannot put “*let’s think step by step*” to warm-start the assistant’s response). All prompts are given in our Huggingface repo, including the model output and what our answer parser extracted as the answer.

Experiments were conducted either by invoking APIs or by running open-source models on our own hardware, mostly on a machine with 8 A40s or 4 Quadro RTX 8000s. All locally hosted models were

⁵These systems use more compute than direct answer, and there is not a clear comparison to be made here. Moreover, our anecdotal coverage of these methods shows that they are most used for math, coding, and logic settings, for which we already have high representation among reported CoT methods.

Table 2: Categories and their descriptions for the meta-analysis.

Category	Description
Symbolic and algorithmic	Tasks involving symbol manipulation which can be solved by executing a program. This includes entity tracking datasets (e.g., SCONE, Coin Flip) and algorithmic tasks (e.g., BBH word sorting or finding shortest paths in a graph).
Math	Tasks requiring mathematical reasoning, from grade-school math to advanced mathematics, including physics questions.
Logical reasoning	Tasks designed to test for logical reasoning, whether deductive (Saparov & He, 2023, PrOntoQA), inductive (Bowen et al., 2024) or analogical (Ma et al., 2024) reasoning, including syllogisms and logical puzzles.
Commonsense reasoning	Datasets designed to test for commonsense knowledge and reasoning, i.e., world knowledge that most people would have, rather than specialized expert-level knowledge in a discipline acquired after years of study.
Encyclopedic knowledge	Tasks requiring expert-level in-depth knowledge beyond mere commonsense, usually in an open-book setting.
Spatial and temporal reasoning	Datasets designed to test for an understanding of space and spatial relations (e.g., navigation) or reasoning involving time and sequences over time.
Multi-hop QA	Questions involving the composition of multiple steps of reasoning in order to arrive at an answer, such as “What is the capital of the country whose scientist discovered penicillin?”
Context-aware QA	Tasks such as closed-book QA and reading comprehension involving reasoning about a given text in context. The context is often a short passage, but could also take the form of a knowledge graph (KBQA) or a table. This category also includes information extraction tasks, such as NER or relation extraction.
Entailment	Tasks involving establishing the inferential relation between two texts, prototypically NLI, but also including fact verification.
Text classification	Tasks involving the classification of a text into a small set of categories, such a topic or sentiment classification, but also involving tasks such as hate speech detection and misinformation detection.
Generation	Tasks involving text generation, including machine translation, dialogue, question generation, as well as code generation. Tasks such as SQL execution (Lei et al., 2024) or systematic transformations of data (e.g., SCAN (Lake & Baroni, 2018)) are excluded because they can be solved by executing a program.
Meta-linguistic	Tasks probing for models’ knowledge of linguistics, such as identifying the main subject of a sentence or solving linguistic puzzles.
Mixed datasets	Datasets containing a variety of tasks, such as BIG-Bench Hard (BBH) or MMLU.
Other	Tasks which did not fit in any of the other categories, such as evaluating AI safety, eliciting models’ verbalized confidence, or melody retrieval.

hosted with vLLM. All parameters given to the vLLM API endpoint are given in the Huggingface repo as well.

D OTHER CoT PROMPT VARIANTS

D.1 TESTING PERFORMANCE VOLATILITY ACROSS PROMPTS

To test the impact of prompt choice on performance, we performed our zero-shot experiment on Llama 3.1 8B with 7 different datasets and 4 different zero-shot CoT prompting strategies common in the literature (Kojima et al., 2022; Wang et al., 2023b; Zhou et al., 2023b; Yang et al., 2024).